

CHELLU UDAY KIRAN

AWS Solutions Architect | AWS Cloud Enthusiast | Aspiring Cloud Engineer |

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Professional Summary

AWS Cloud enthusiast with hands-on knowledge of AWS Solution Architect concepts, networking fundamentals, and Linux operating systems. Passionate about building secure, scalable, and reliable cloud solutions while applying cloud security and cost optimization best practices.

Education

Pragati Engineering College

B.Tech - Electronics and Communication Engineering

CGPA: 8.3

Narayana Junior College

Intermediate (MPC)

GPA: 9.3

Narayana E-Techno School

Secondary Education

GPA: 10

Technical Skills

Cloud Platforms: AWS

AWS Services: EC2, S3, IAM, VPC, Route53, CloudFront, RDS, ELB, Auto Scaling, CloudWatch, CloudFormation, CloudTrail, EBS, EFS, Lambda, Cloud Security

Architecture Principles: AWS Well-Architected Framework, Multi-tier Architecture, Serverless Architecture, Microservices, High Availability & Fault Tolerance, Disaster Recovery (RTO/RPO), Cost Optimization

Networking: TCP/IP, OSI Model, DNS, HTTP/HTTPS, DHCP, Subnetting, CIDR, VPC Peering, VPN, Direct Connect concepts, Security Groups, Internet Gateway

Operating Systems: Linux (Ubuntu, Kali), Windows

Programming: Python, Bash, Shell Scripting

Version Control: Git, GitHub

Tools: AWS CLI, Wireshark, VS Code, PuTTY

Projects

Static Website Hosting on AWS (S3 + CloudFront + Route 53)

- Hosted a static website on Amazon S3 with static website hosting enabled and bucket policies for secure public access; integrated CloudFront CDN for global low-latency content delivery
- Configured SSL/TLS using AWS Certificate Manager (ACM) and Route 53 for custom domain routing and DNS management
- Implemented S3 versioning and lifecycle policies for cost-optimized storage management
- Improved global content delivery performance using Amazon CloudFront edge locations

Multi-Tier Web Application Architecture (AWS)

- Designed a scalable three-tier architecture consisting of web, application, and database layers.
- Configured VPC with public and private subnets.
- Integrated Application Load Balancer and Auto Scaling Groups.
- Utilized RDS for managed database services.
- Implemented security groups and network isolation.

Linux Process & Thread Simulator

- Developed a Linux-based application in C to simulate process creation, multithreading, and concurrent task execution using POSIX threads and fork system calls
- Implemented Inter-Process Communication (IPC) mechanisms including pipes, shared memory, and message queues for inter-process data exchange

- Applied thread synchronization primitives (mutexes, semaphores) to prevent race conditions and ensure data integrity in concurrent operations

Serverless Event-Driven Architecture with AWS Lambda

- Built a serverless data processing pipeline using AWS Lambda, S3 event triggers, and DynamoDB for real-time file processing and metadata storage
- Configured Lambda functions with appropriate IAM execution roles, VPC placement, and environment variables for secure and isolated execution
- Implemented SNS and SQS for decoupled, asynchronous message processing between microservice components
- Utilized CloudFormation templates for Infrastructure as Code (IaC), enabling repeatable and version-controlled deployments across environments

Experience

AWS Cloud Architecture Hands-On Lab

Hands-On Practice

- Designed and deployed 10+ AWS cloud architectures covering high availability, security, scalability, and disaster recovery scenarios.
- Practiced IAM policies, VPC networking, storage solutions, monitoring, and security implementation.
- Gained practical understanding of designing secure and scalable systems.
- Provisioned AWS resources using AWS Console and AWS CLI.

Cybersecurity & Network Security Lab

VMware Workstation

- Configured vulnerable test environments using DVWA and OWASP Juice Shop to practice identifying OWASP Top 10 vulnerabilities including SQL Injection and XSS
- Conducted network reconnaissance and vulnerability assessments using Nmap, Wireshark, and Burp Suite in controlled lab environments
- Analyzed packet captures to understand TCP handshake, DNS resolution, and HTTP request/response flows for network troubleshooting

Achievements & Certifications

- AWS Solutions Architect Learning Path
- Cisco CCNA: Introduction to Networks
- ServiceNow Certified System Administrator (CSA)
- ServiceNow Certified Application Developer (CAD)